

Bill of Materials — VBF-T3R1-BOM-001

Sheet: BOM

| Component | Material | Thickness [um] | Porosity | Density [kg/m3] | Area [m2] | Mass [g] | Source

1 | Positive electrode (NMC811) | Positive electrode | 40 | 0.42 | 3699 | 0.1027 | 7.772 | Chen2020 set

2 | Negative electrode (graphite) | Negative electrode | 52 | 0.35 | 2060 | 0.1027 | 5.752 | Chen2020 set

3 | Separator (polyolefin) | Separator | 12 | 0.47 | 1548 | 0.1027 | 0.259 | Chen2020 set

4 | Positive current collector (Al) | Positive current collector | 16 | - | 2702 | 0.1027 | 4.437 | Chen2020 set

5 | Negative current collector (Cu) | Negative current collector | 12 | - | 8933 | 0.1027 | 11.042 | Chen2020 set

| Electrolyte | 1M LiPF6 EC/EMC (kappa 1.5 S/m, D 2.5e-10, t+ 0.35) | - | - | - | - | not provided (density absent from parameter set; contract excludes) | p_v12.json / calc-energy note

| TOTAL (active stack) | | 132 | | | 0.1027 | 29.26 | calc-energy mass_kg

Design values | | particle radii: pos 2.0 um / neg 2.5 um | | | | | p_v12.json

Cooling | h = 20 W/m2/K | cell-to-ambient | | | | | p_v12.json

Voltage window | 2.5 - 4.2 V | | | | | Chen2020 set